Algorithmic Treatment

Solving balance of linear momentum requires evaluation of constitutive law:

$$\underline{\sigma} = \underline{\sigma}(\underline{\varepsilon}^e) \rightsquigarrow \underline{\varepsilon}^e = \underline{\varepsilon} - \underline{\varepsilon}^p$$

$$\rightsquigarrow \underline{\varepsilon}^p \text{ from } \dot{\underline{\varepsilon}}^p$$

Time discretization and numerical integration

$$[0, t] = \bigcup_{n=1}^{N} [t_n, t_{n+1}] \rightsquigarrow \Delta t = t_{n+1} - t_n$$

in current time interval, values at t_n are known, i.e.

$$\underset{\sim}{\varepsilon}_n(\underset{\sim}{x}),\ \underset{\sim}{\varepsilon}^p_n(\underset{\sim}{x}),\ \alpha_n(\underset{\sim}{x}),\ \underset{\sim}{u}_n(\underset{\sim}{x})$$

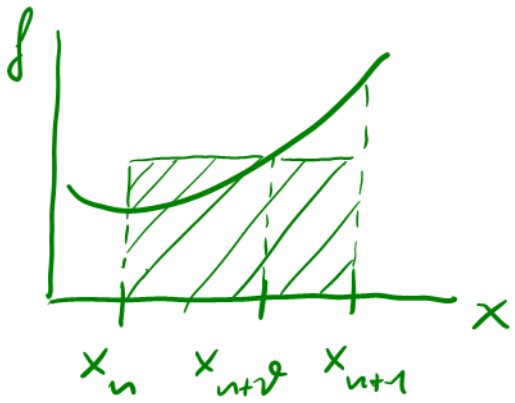
$\underset{\sim}{u}_{n+1}(\underset{\sim}{x})$ can be computed via FEM + Newton-Raphson (spatial discretization), therein we need to compute

α_{n+1}, $\underline{\varepsilon}^p_{n+1}$ (from time integration of evolution eq.)

$\hookrightarrow \underline{\varepsilon}^e_{n+1} \rightsquigarrow \underline{\sigma}_{n+1}$

Time integration:

$$\int_{t_n}^{t_{n+1}} f(x(t))\,dt \approx \underbrace{\Delta t\, f(x_{n+\vartheta})}_{\text{generalized midpoint rule}}$$



with $x_{n+\vartheta} = \vartheta x_{n+1} + (1-\vartheta) x_n$

(1) Explicit Euler: $\vartheta = 0$

(2) Midpoint rule: $\vartheta = \frac{1}{2}$

(3) Implicit Euler: $\vartheta = 1$

- second-order accuracy only for $\vartheta = \frac{1}{2}$,
- unconditionally stable for $\vartheta \geq \frac{1}{2}$

Projection algorithm (for calculation of current $\underline{\varepsilon}^p$)

(i) Elastic sub-problem (predictor step $\hat{=}$ trial step)

Assume in current time step that $\dot{\underline{\varepsilon}}^e = \dot{\underline{\varepsilon}}$

$$\dot{\underline{\varepsilon}}^e = \dot{\underline{\varepsilon}} \;\Big|\; \int(\dots)dt$$

$$\Leftrightarrow \quad \underline{\varepsilon}^e_{n+1} - \underline{\varepsilon}^e_n = \int_{t_n}^{t_{n+1}} \dot{\underline{\varepsilon}}\, dt$$

$$\Leftrightarrow \quad \underline{\varepsilon}^e_{n+1} = \underline{\varepsilon}^e_n + \underline{\varepsilon}_{n+1} - \underline{\varepsilon}_n = \underline{\varepsilon}_{n+1} - (\underline{\varepsilon}_n - \underline{\varepsilon}^e_n)$$

$$\Leftrightarrow \quad \underline{\varepsilon}^{e,\text{trial}}_{n+1} = \underline{\varepsilon}_{n+1} - \underline{\varepsilon}^p_n \qquad (1)$$

$$\rightsquigarrow \underline{\sigma}^{\text{trial}} := \underline{\sigma}\Big|_{\underline{\varepsilon}^e = \underline{\varepsilon}^{e,\text{trial}}_{n+1}} \quad ; \quad \beta^{\text{trial}} = \beta\Big|_{\alpha_n} \qquad (2)$$

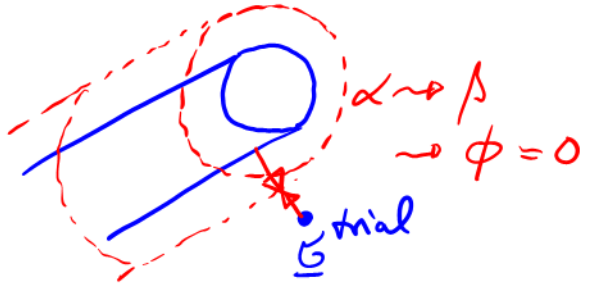
$$\rightsquigarrow \phi^{\text{trial}} := \phi\Big|_{\underline{\sigma}^{\text{trial}},\, \beta^{\text{trial}}} = \|\text{dev}\,\underline{\sigma}^{\text{trial}}\| - \sqrt{\tfrac{2}{3}}\left(y_0 + \beta^{\text{trial}}\right) \qquad (3)$$

- if $\phi^{\text{trial}} \leq 0$, then trial step is indeed elastic

  $\rightsquigarrow \underline{\sigma} = \underline{\sigma}^{\text{trial}}$ ✓

- if $\phi^{\text{trial}} > 0$, then correction required

(ii) Plastic sub-problem (corrector step)



Direction of correction is normal to ϕ, i.e.

$$\underline{n} = \frac{\partial \phi}{\partial \underline{\sigma}} = \underline{n}\Big|_{trial} =: \underline{n}_{trial}$$

- $$\underline{\varepsilon}^e_{n+1} = \underline{\varepsilon}_{n+1} - \underbrace{\underline{\varepsilon}^p_{n+1}}_{=?}$$

$$\rightsquigarrow \int_{t_n}^{t_{n+1}} \dot{\underline{\varepsilon}}^p \, dt = \underline{\varepsilon}^p_{n+1} - \underline{\varepsilon}^p_n = \underbrace{\int_{t_n}^{t_{n+1}} \lambda \, \underline{n} \, dt}$$

$$\approx \Delta t \, \lambda_{n+1} \, \underline{n}_{n+1} \quad \text{(implicit Euler)}$$

$$\Leftrightarrow \underline{\varepsilon}^p_{n+1} = \underline{\varepsilon}^p_n + \underbrace{\Delta t \, \lambda_{n+1} \, \underline{n}_{n+1}}_{=: \Delta \underline{\varepsilon}^p_{n+1}} \quad \Big| \text{ linear in } \lambda_{n+1} \qquad (4)$$

- $$\dot{\alpha} = \lambda \sqrt{\tfrac{2}{3}} \Leftrightarrow \alpha_{n+1} = \alpha_n + \int_{t_n}^{t_{n+1}} \sqrt{\tfrac{2}{3}} \lambda \, dt$$

$$\approx \alpha_n + \sqrt{\tfrac{2}{3}} \lambda_{n+1} \Delta t \quad \Big| \text{ lin. in } \lambda_{n+1} \qquad (5)$$

$\Rightarrow \underline{\varepsilon}^p_{n+1}$ and α_{n+1} depend on λ_{n+1} !

For evolution of plasticity, we have $\phi = 0$

$\rightsquigarrow$ compute λ from $\phi = \underbrace{\|\mathrm{dev}\,\underline{\sigma}\|}_{\text{depends on } \underline{\varepsilon}^p(\lambda),\ \alpha(\lambda)} - \sqrt{\frac{2}{3}}\left(y_0 + \beta(\alpha)\right) = 0$

$\Rightarrow$ if $\|\mathrm{dev}\,\sigma\|$ linear in $\underline{\varepsilon}^p$ and α $\rightsquigarrow$ linear in λ

and $\beta(\alpha)$ linear in α $\rightsquigarrow$ linear in λ

$\Rightarrow$ ϕ linear in λ $\rightsquigarrow$ can be solved analytically

otherwise numerical solution, e.g. via local Newton

$\rightsquigarrow$ "Multi-level Newton"

Model problem: Linear von Mises hardening

- $\psi = \psi^e + \psi^p$ with $\psi^e = \frac{1}{2}\kappa (\mathrm{tr}\,\underline{\varepsilon}^e)^2 + \mu \|\mathrm{dev}\,\underline{\varepsilon}^e\|^2$

$$\psi^p = \frac{1}{2} H \alpha^2$$ hardening modulus

$$\underline{\sigma} = \kappa (\mathrm{tr}\,\underline{\varepsilon}^e)\underline{1} + 2\mu\,\mathrm{dev}\,\underline{\varepsilon}^e$$

$$\underline{\mathbb{C}}^e = \frac{\partial \underline{\sigma}}{\partial \underline{\varepsilon}^e} = \underbrace{\kappa\, \underline{1}\otimes\underline{1}}_{\underline{\mathbb{C}}^{e,\mathrm{vol}}} + \underbrace{2\mu\, \underline{\mathbb{P}}}_{\underline{\mathbb{C}}^{e,\mathrm{iso}}}$$

$$\left.\begin{aligned} \underline{\sigma}^{\mathrm{vol}} &= \underline{\mathbb{C}}^{e,\mathrm{vol}} : \underline{\varepsilon}^e \\ \mathrm{dev}\,\underline{\sigma} &= \underline{\mathbb{C}}^{e,\mathrm{iso}} : \underline{\varepsilon}^e \\ \hookrightarrow \underline{\sigma} &= \underline{\sigma}^{\mathrm{vol}} + \mathrm{dev}\,\underline{\sigma} \end{aligned}\right\}$$

- Hardening: $\beta = \frac{\partial \psi}{\partial \alpha} = H\alpha$
- Yield criterion: $\phi = \|\mathrm{dev}\,\underline{\sigma}\| - \sqrt{\frac{2}{3}}(y_0 + \beta) \leq 0$
- Flow rule: $\dot{\underline{\varepsilon}}^p = \lambda \frac{\partial \phi}{\partial \underline{\sigma}} = \lambda \underline{n}$
- Normal tensor: $\underline{n} = \frac{\partial \phi}{\partial \underline{\sigma}} = \underbrace{\frac{\partial \|\mathrm{dev}\,\underline{\sigma}\|}{\partial\, \mathrm{dev}\,\underline{\sigma}}}_{= \frac{\mathrm{dev}\,\underline{\sigma}}{\|\mathrm{dev}\,\underline{\sigma}\|}} : \underbrace{\frac{\partial\, \mathrm{dev}\,\underline{\sigma}}{\partial \underline{\sigma}}}_{\underline{\mathbb{P}}} = \frac{\mathrm{dev}\,\underline{\sigma}}{\|\mathrm{dev}\,\underline{\sigma}\|}$
- Evolution of internal variable $\dot{\alpha} = -\lambda \frac{\partial \phi}{\partial \beta} = \sqrt{\frac{2}{3}}\lambda$
- Kuhn-Tucker-conditions: $\lambda \geq 0,\ \phi \leq 0,\ \lambda\phi = 0$

(i) Predictor step:

(1): $\underline{\varepsilon}^{e,trial}_{n+1} = \underline{\varepsilon}_{n+1} - \underline{\varepsilon}^{p}_{n}$

(2): $\underline{\sigma}^{trial} = \underline{\sigma}^{vol,trial} + dev\,\underline{\sigma}^{trial} = \kappa\, tr\,\underline{\varepsilon}^{e,trial}_{n+1}\,\underline{1} + 2\mu\, dev\,\underline{\varepsilon}^{e,trial}_{n+1}$

$\alpha^{trial}_{n+1} = \alpha_n \rightsquigarrow \beta^{trial} = H\alpha_n$

(3): $\phi^{trial} = \|dev\,\underline{\sigma}^{trial}\| - \sqrt{\frac{2}{3}}(y_0 + H\alpha_n)$ (6)

if $\phi^{trial} \leq 0$, then $\underline{\sigma} = \underline{\sigma}^{trial}$, $\underline{\varepsilon}^{p}_{n+1} = \underline{\varepsilon}^{p}_{n}$, $\alpha_{n+1} = \alpha_n$

$\underline{\mathbb{C}} = \underline{\mathbb{C}}^{e,vol} + \underline{\mathbb{C}}^{e,iso}$

if $\phi^{trial} > 0$, then ...

(ii) Corrector step:

$$dev\,\underline{\sigma} = \underline{\mathbb{C}}^{e,iso} : \underline{\varepsilon}^{e}_{n+1} \overset{(4)}{=} \underline{\mathbb{C}}^{e,iso} : \Big(\underbrace{\underline{\varepsilon}_{n+1} - \underline{\varepsilon}^{p}_{n}}_{=\underline{\varepsilon}^{e,trial}_{n+1}\ (1)} - \Delta t\,\lambda_{n+1}\,\underline{n}_{n+1}\Big)$$

$= \underline{\varepsilon}_{n+1} - \underline{\varepsilon}^{p}_{n+1} = \underline{\varepsilon}_{n+1} - \underline{\varepsilon}^{p}_{n} - \Delta\underline{\varepsilon}^{p}_{n+1}$

$$\Leftrightarrow\ dev\,\underline{\sigma} = \underline{\mathbb{C}}^{e,iso} : \underline{\varepsilon}^{e,trial}_{n+1} - 2\mu\,\underline{\mathbb{P}} : \lambda_{n+1}\Delta t\,\underline{n}_{n+1} \quad \Big| \ \underline{\mathbb{P}} : \underline{n} = \underline{n} \text{ da } \underline{n} = \frac{dev\,\underline{\sigma}}{\|dev\,\underline{\sigma}\|}$$

$$\Leftrightarrow\ dev\,\underline{\sigma} = dev\,\underline{\sigma}^{trial} - 2\mu\,\lambda_{n+1}\Delta t\,\underline{n}_{n+1} \qquad (7)$$

$$\Leftrightarrow \frac{\|dev\,\underline{\sigma}\|}{\|dev\,\underline{\sigma}\|}\,dev\,\underline{\sigma} = \frac{\|dev\,\underline{\sigma}^{trial}\|}{\|dev\,\underline{\sigma}^{trial}\|}\,dev\,\underline{\sigma}^{trial} - 2\mu\,\lambda_{n+1}\Delta t\,\underline{n}_{n+1}$$

$$\Leftrightarrow \|dev\,\underline{\sigma}\|\,\underline{n}_{n+1} = \|dev\,\underline{\sigma}^{trial}\|\,\underbrace{\underline{n}^{trial}}_{=\underline{n}_{n+1}} - 2\mu\,\lambda_{n+1}\Delta t\,\underline{n}_{n+1}$$

$$\rightsquigarrow \|dev\,\underline{\sigma}\| = \underbrace{\|dev\,\underline{\sigma}^{trial}\|}_{\text{independent from } \lambda_{n+1}} - 2\mu\,\lambda_{n+1}\Delta t \quad \Big\{ \text{lin. in } \lambda_{n+1}$$

insert in yield criterion:

$$\phi_{n+1} = \| \operatorname{dev} \underline{\sigma}^{\text{trial}}_{n+1} \| - 2\mu \lambda_{n+1} \Delta t - \sqrt{\tfrac{2}{3}} \left(y_0 + H \alpha_{n+1} \right)$$

$$\overset{(5)}{=} \alpha_n + \sqrt{\tfrac{2}{3}} \lambda_{n+1} \Delta t$$

$$= \underbrace{\| \operatorname{dev} \underline{\sigma}^{\text{trial}} \| - \sqrt{\tfrac{2}{3}} \left(y_0 + H \alpha_n \right)}_{\overset{(6)}{=} \phi^{\text{trial}}} - \left(2\mu + \tfrac{2}{3} H \right) \Delta t \, \lambda_{n+1}$$

$$= \phi^{\text{trial}} - \left(2\mu + \tfrac{2}{3} H \right) \Delta t \, \lambda_{n+1} \quad \leftarrow \text{lin. in } \lambda_{n+1}$$

plastic evolution for $\phi_{n+1} = 0$

$$\phi^{\text{trial}} - \left(2\mu + \tfrac{2}{3} H \right) \Delta t \, \lambda_{n+1} \overset{!}{=} 0$$

$$\Leftrightarrow \lambda_{n+1} = \frac{\phi^{\text{trial}}}{\left(\tfrac{2}{3} H + 2\mu \right) \Delta t} \qquad (8)$$

- Stresses:

$$(7): \operatorname{dev} \underline{\sigma} = \operatorname{dev} \underline{\sigma}^{\text{trial}} - 2\mu \lambda_{n+1} \Delta t \, \underline{n} = \bar{c} \operatorname{dev} \underline{\sigma}^{\text{trial}}$$

$$\text{with } \bar{c} = 1 - \frac{2\mu \lambda_{n+1} \Delta t}{\| \operatorname{dev} \underline{\sigma}^{\text{trial}} \|} = 1 - \frac{2\mu \, \phi^{\text{trial}}}{\left(\tfrac{2}{3} H + 2\mu \right) \| \operatorname{dev} \underline{\sigma}^{\text{trial}} \|}$$

$$\Rightarrow \underline{\sigma} = \lambda \left(\operatorname{tr} \underline{\varepsilon}^e \right) \underline{1} + \operatorname{dev} \underline{\sigma}$$

- History variables:

$$(4): \underline{\varepsilon}^p_{n+1} = \underline{\varepsilon}^p_n + \Delta t \lambda_{n+1} \underline{n}_{n+1}$$

$$(5): \alpha_{n+1} = \alpha_n + \sqrt{\tfrac{2}{3}} \lambda_{n+1} \Delta t$$

- Elasto-plastic tangent moduli:

- consider stress linearization $\Delta \underline{\sigma} = \underbrace{\frac{\partial \underline{\sigma}}{\partial \underline{\varepsilon}}}_{\underline{\mathbb{C}}^{ep} = ?} : \Delta \underline{\varepsilon}$

$$\Delta \underline{\sigma} = \underbrace{\Delta \underline{\sigma}^{vol}}_{\text{purely elastic} = \underline{\mathbb{C}}^{e,vol} : \Delta \underline{\varepsilon}} + \Delta \underline{\sigma}^{iso}$$

- Deviatoric part

$$\Delta \underline{\sigma}^{iso} = \underbrace{\frac{\partial \operatorname{dev} \underline{\sigma}}{\partial \underline{\varepsilon}^e}}_{\underline{\mathbb{C}}^{e,iso} = 2\mu \underline{\mathbb{P}}} : \underbrace{\frac{\partial \underline{\varepsilon}^e}{\partial \underline{\varepsilon}}}_{?} : \Delta \underline{\varepsilon} \qquad (9)$$

$\hookrightarrow \underline{\varepsilon}^e = \underline{\varepsilon} - \underline{\varepsilon}^p$

$\overset{(4)}{=} \underline{\varepsilon} - \underline{\varepsilon}^p_n - \lambda \Delta t \underline{n}^{trial}$

$\overset{(8)}{=} \underline{\varepsilon} - \underline{\varepsilon}^p - \frac{\phi^{trial}}{(\frac{2}{3}H + 2\mu)} \underline{n}^{trial}$

$$\rightsquigarrow \frac{\partial \underline{\varepsilon}^e}{\partial \underline{\varepsilon}} = \underbrace{\frac{\partial \underline{\varepsilon}}{\partial \underline{\varepsilon}}}_{\underline{\mathbb{I}}} - \underbrace{\frac{\partial \underline{\varepsilon}^p_n}{\partial \underline{\varepsilon}}}_{\underline{\mathbb{O}}} - \frac{1}{(\frac{2}{3}H + 2\mu)} \left(\underbrace{\frac{\partial \phi^{trial}}{\partial \underline{\varepsilon}}}_{(a)} \otimes \underline{n}^{trial} + \phi^{trial} \underbrace{\frac{\partial \underline{n}^{trial}}{\partial \underline{\varepsilon}}}_{(b)} \right)$$

($\underline{\varepsilon}^p_n$ does not depend on current $\underline{\varepsilon}$)

$$(a): \frac{\partial \phi^{trial}}{\partial \underline{\varepsilon}} = \frac{\partial \left[\| \operatorname{dev} \underline{\sigma}^{trial} \| - \sqrt{\tfrac{2}{3}} (y_0 + H \alpha_n) \right]}{\partial \underline{\varepsilon}}$$

$$= \frac{\partial \| \operatorname{dev} \underline{\sigma}^{trial} \|}{\partial \operatorname{dev} \underline{\sigma}^{trial}} : \frac{\partial \operatorname{dev} \underline{\sigma}^{trial}}{\partial \underline{\varepsilon}}$$

$$= \underbrace{\frac{\operatorname{dev} \underline{\sigma}^{trial}}{\| \operatorname{dev} \underline{\sigma}^{trial} \|}}_{\underline{n}^{trial}} : \underbrace{\frac{\partial 2\mu \operatorname{dev} \underline{\varepsilon}^{e,trial}}{\partial \underline{\varepsilon}^{e,trial}}}_{2\mu \underline{\mathbb{P}}} : \underbrace{\frac{\partial \underline{\varepsilon}^{e,trial}}{\partial \underline{\varepsilon}}}_{\underline{\mathbb{I}}}$$

$$= 2\mu \underline{n}^{trial}$$

(b): $$\underline{n}^{trial} = \frac{dev\,\underline{\sigma}^{trial}}{\|dev\,\underline{\sigma}^{trial}\|} = \frac{2\mu\, dev\,\underline{\varepsilon}^{e,trial}}{\|2\mu\, dev\,\underline{\varepsilon}^{e,trial}\|} \qquad (*)$$

$$\rightsquigarrow \frac{\partial \underline{n}^{trial}}{\partial \underline{\varepsilon}} = \frac{\partial \underline{n}^{trial}}{\partial \underline{\varepsilon}^{e,trial}} : \underbrace{\frac{\partial \underline{\varepsilon}^{e,trial}}{\partial \underline{\varepsilon}}}_{\underline{\underline{\mathbb{I}}}}$$

$$= \underbrace{\frac{\partial \|dev\,\underline{\varepsilon}^{e,trial}\|^{-1}}{\partial \underline{\varepsilon}^{e,trial}}}_{-\frac{1}{\|dev\,\underline{\varepsilon}^{e,trial}\|^3}\, dev\,\underline{\varepsilon}^{e,trial}} \otimes dev\,\underline{\varepsilon}^{e,trial} + \frac{1}{\|dev\,\underline{\varepsilon}^{e,trial}\|} \underbrace{\frac{\partial dev\,\underline{\varepsilon}^{e,trial}}{\partial \underline{\varepsilon}^{e,trial}}}_{\underline{\underline{\mathbb{P}}}}$$

$$= \frac{1}{\|dev\,\underline{\varepsilon}^{e,trial}\|}\left(\underline{\underline{\mathbb{P}}} - \frac{dev\,\underline{\varepsilon}^{e,trial}}{\|dev\,\underline{\varepsilon}^{e,trial}\|} \otimes \frac{dev\,\underline{\varepsilon}^{e,trial}}{\|dev\,\underline{\varepsilon}^{e,trial}\|}\right)$$

$$\overset{(*)}{=} \frac{1}{\|dev\,\underline{\varepsilon}^{e,trial}\|}\left(\underline{\underline{\mathbb{P}}} - \underline{n}^{trial} \otimes \underline{n}^{trial}\right)$$

einsetzen:

$$\frac{\partial \underline{\varepsilon}^{e}}{\partial \underline{\varepsilon}} = \underline{\underline{\mathbb{I}}} - \frac{1}{(\frac{2}{3}H + 2\mu)} 2\mu\, \underline{n}^{trial} \otimes \underline{n}^{trial} - \underbrace{\frac{\phi^{trial}}{(\frac{2}{3}H + 2\mu)\|dev\,\underline{\varepsilon}^{e,trial}\|}}_{=:c}\left(\underline{\underline{\mathbb{P}}} - \underline{n}^{trial} \otimes \underline{n}^{trial}\right)$$

in (9):

$$\Delta\underline{\sigma}^{iso} = 2\mu\underline{\underline{\mathbb{P}}} : \underbrace{\left[\underline{\underline{\mathbb{I}}} - \frac{2\mu}{(\frac{2}{3}H + 2\mu)} \underline{n}^{trial} \otimes \underline{n}^{trial} - c\,\underline{\underline{\mathbb{P}}} + c\, \underline{n}^{trial} \otimes \underline{n}^{trial}\right]}_{\underline{\underline{\mathbb{C}}}^{ep,iso}} : \Delta\underline{\varepsilon}$$

$$\underline{\underline{\mathbb{C}}}^{ep,iso} = 2\mu\underline{\underline{\mathbb{P}}}(1 - c) - 2\mu\left(\frac{\mu}{(\frac{H}{3} + \mu)} + c\right) \underline{n}^{trial} \otimes \underline{n}^{trial}$$

$$\Rightarrow \underline{\underline{\mathbb{C}}}^{ep} = \underline{\underline{\mathbb{C}}}^{e,vol} + \underline{\underline{\mathbb{C}}}^{ep,iso}$$